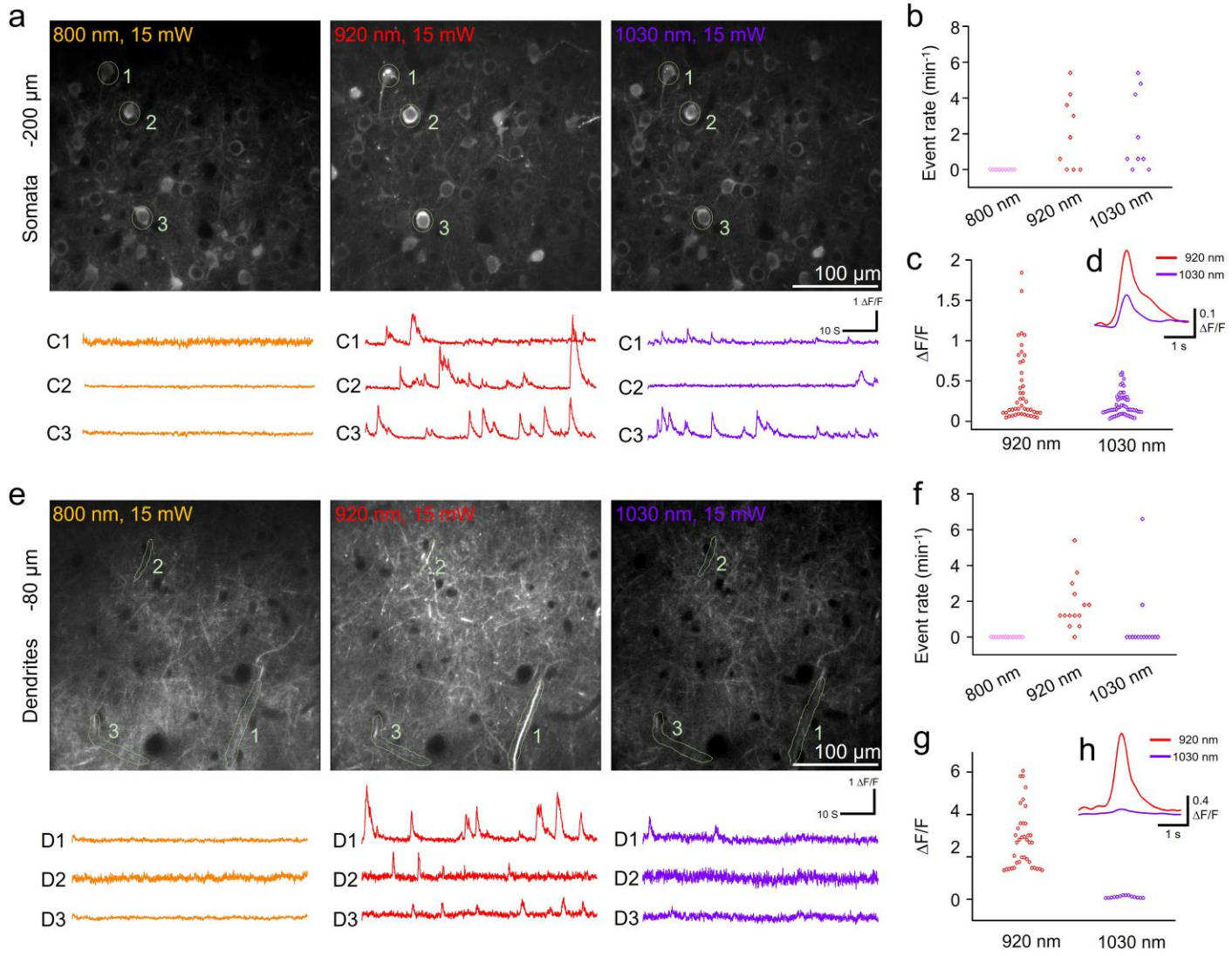
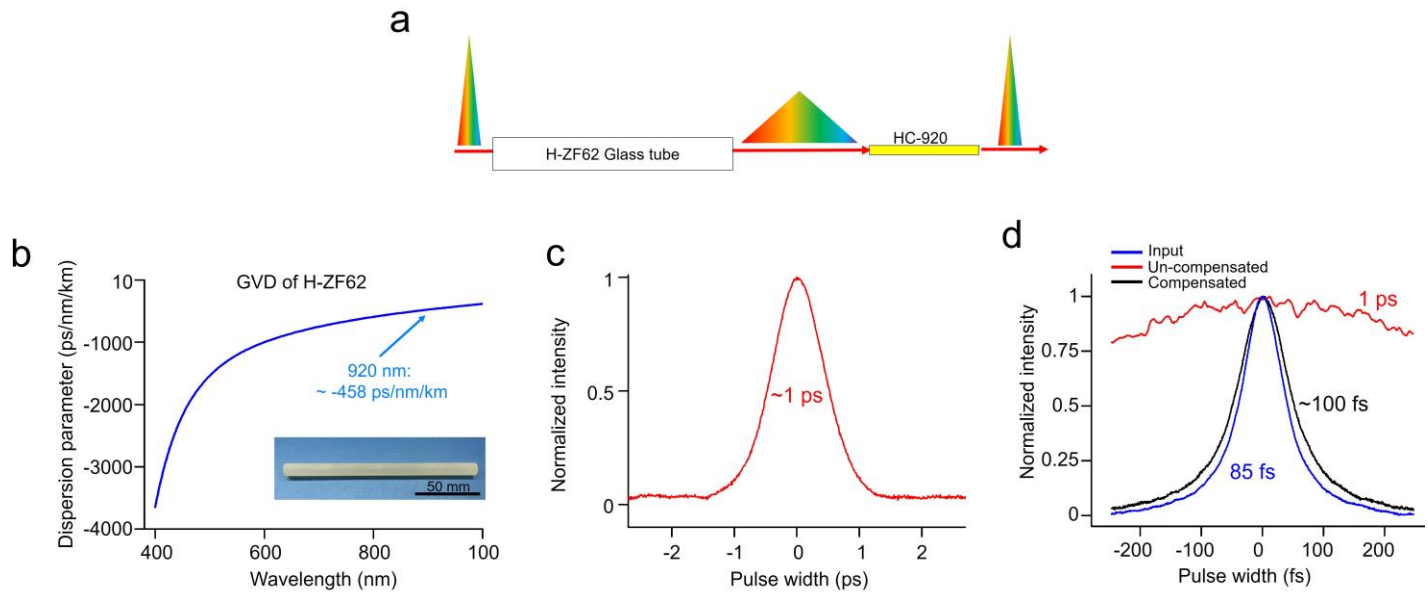


Supplementary Figure 1

Comparison of two-photon excitation efficiency of GCaMP-6f at 800-nm, 920-nm, and 1,030-nm excitation.

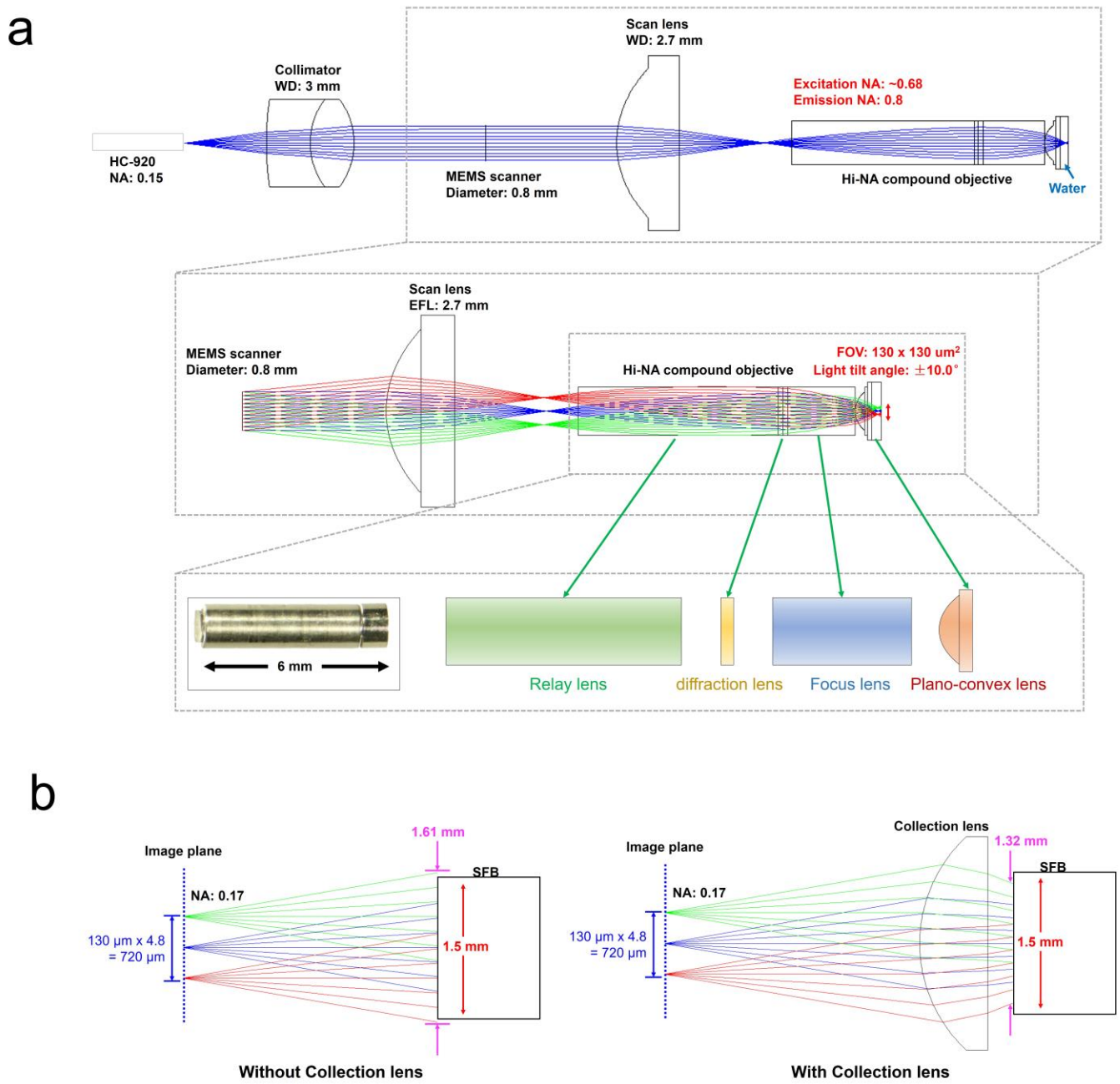
(**a**) Imaging the same sets of neuronal somata in an AAV-transfected mouse expressing GCaMP-6f in V-1 cortex at different excitation wavelengths. (**b**)-(**d**) Frequencies (**b**), amplitudes (**c**), and averaged time courses (**d**) of Ca^{2+} transients from ten somata shown in (**a**). (**e**) As in (**a**), except for imaging neuronal dendrites at a different focal plane. (**f**)-(**h**) Frequencies (**f**), amplitudes (**g**), and average time courses (**h**) of Ca^{2+} transients from ten dendrites shown in (**e**). Imaging parameters are shown in **Supplementary Table S6.**



Supplementary Figure 2

Dispersion compensation of the HC-920.

(**a**) Schematic for compensation of the HC-920 dispersion with an H-ZF62 glass tube. (**b**) Dispersion parameter of the glass H-ZF62. (**c**) Pulse-width of the 920-nm femtosecond laser output from a 1-m HC-920 without dispersion compensation. (**d**) Pulse-width of the 920-nm femtosecond laser before and after a 1-m HC-920 with and without dispersion compensation. See **Supplementary Note 1** for details.

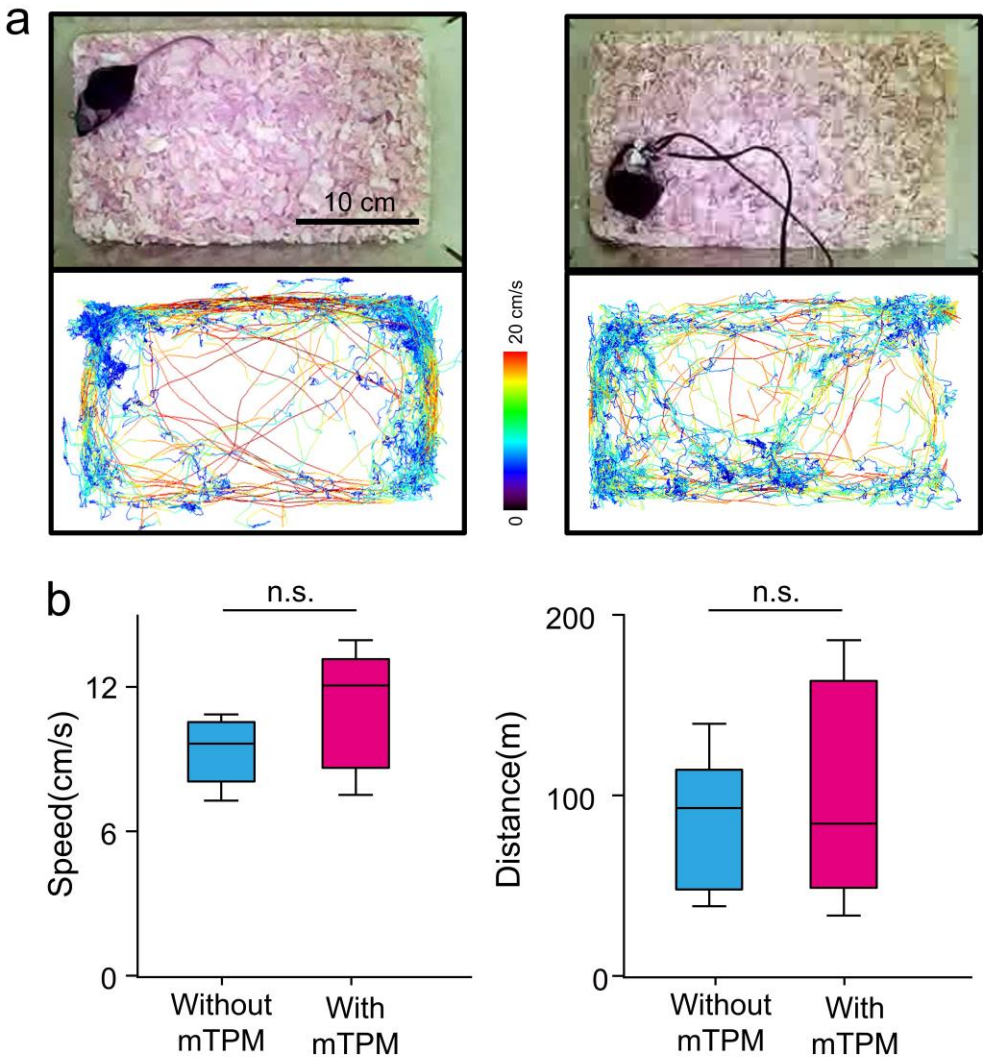


Supplementary Figure 3

Optical design of the FHIRM-TPM.

(**a**) Illumination arm of the FHIRM-TPM. Upper: on-axis excitation light path from the output of the HC-920 to the specimen. Middle: off-axis scanning light path from the surface of the MEMS to the specimen. Lower: schematic of the high NA achromatic miniature compound objective assembly. (**b**) Detection arm of the FHIRM-TPM. Left: without the collection lens, some of the emission light at the edge of a large FOV (130 μm) is lost. Right: The collection lens relays all emission light within the FOV onto the SFB. See details in **Supplementary Note 2.**

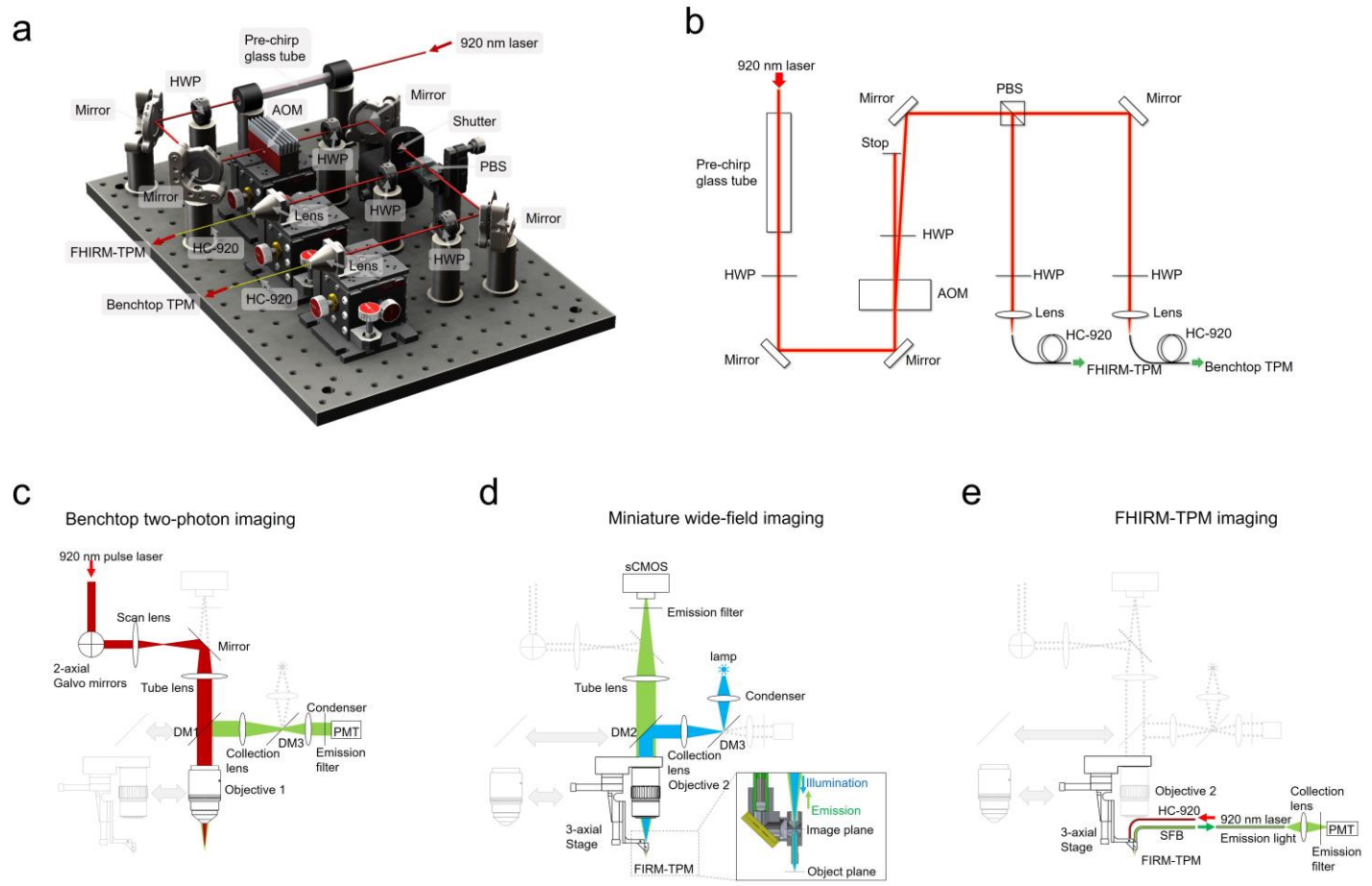
.



Supplementary Figure 4

Possible effect of the carried headpiece on mouse movement.

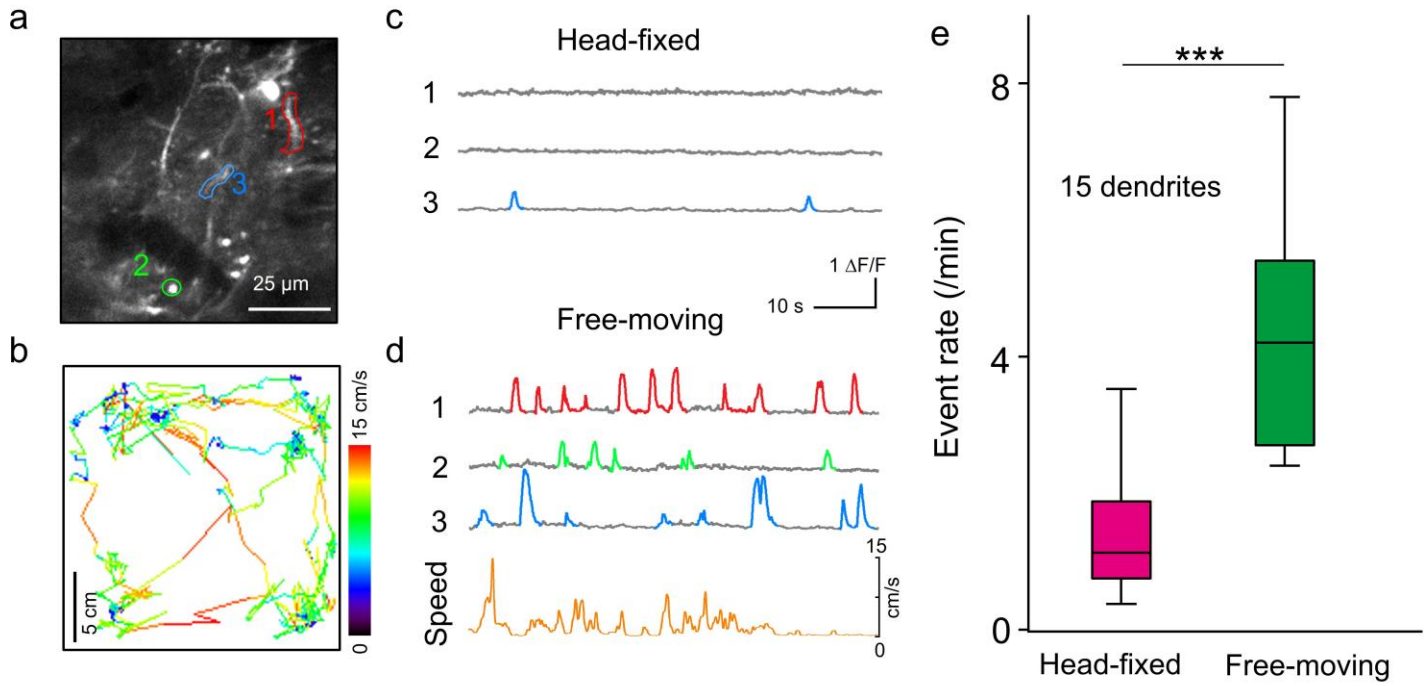
(**a**) Snapshots from videos captured by the camera showing a representative mouse without (left) or with the FHIRM-mTPM mounted on its head (right). Trajectories extracted from these videos are presented in the below. The speed of movement is coded in color. (**b**) Average moving speed (left) and total travel distance (right) of mice with and without the FHIRM-mTPM. Data are presented as box-and-whisker plot from 8 mice with or without FHIRM-TPM. Two-tailed paired t-test. *p* for speed = 0.103. *p* for distance = 0.324.



Supplementary Figure 5

An integrated platform for comparing the performance of benchtop TPM, miniature wide-field microscope, and FHIRM-TPM.

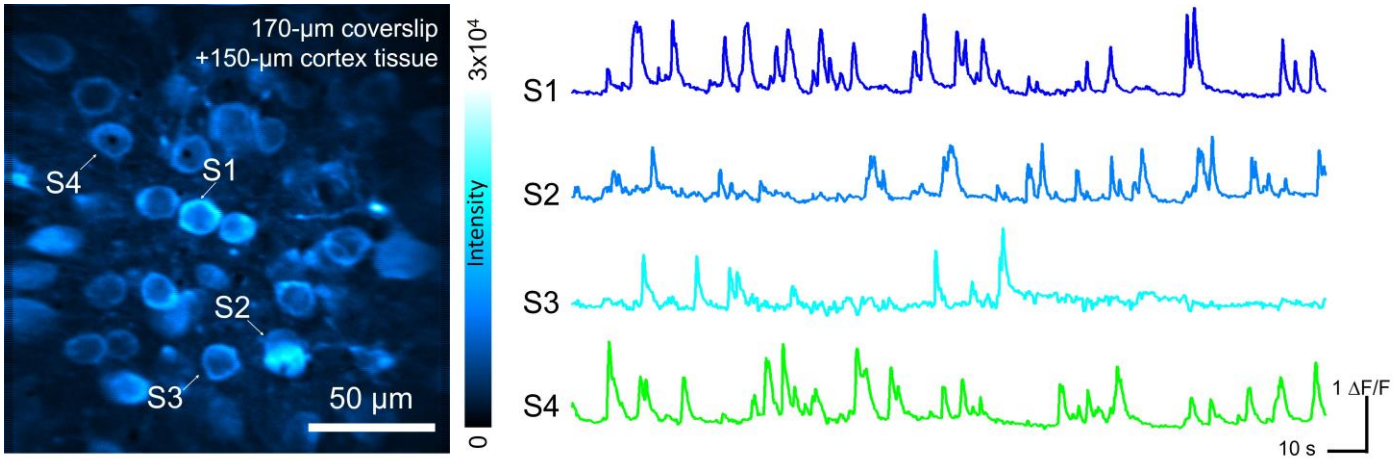
(a) 3D schematic of excitation ligh path for benchtop TPM and FHIRM-TPM. HWP: half wave plate; AOM: acoustic-optics modulator; PBS: polarization beam splitter. **(b)** Optical diagram of the light path shown in (a). **(c)-(e)** Schematic of the light path of the real setup switchable among a benchtop TPM (c, d), a miniature wide-field fluorescence microscope (e, f), and an FHIRM-TPM (g, h). See more details in **Supplementary Note 3.**



Supplementary Figure 6

Comparison of dendrite activities in V1 cortex under head-fixed and freely moving conditions in a mouse in a dark arena.

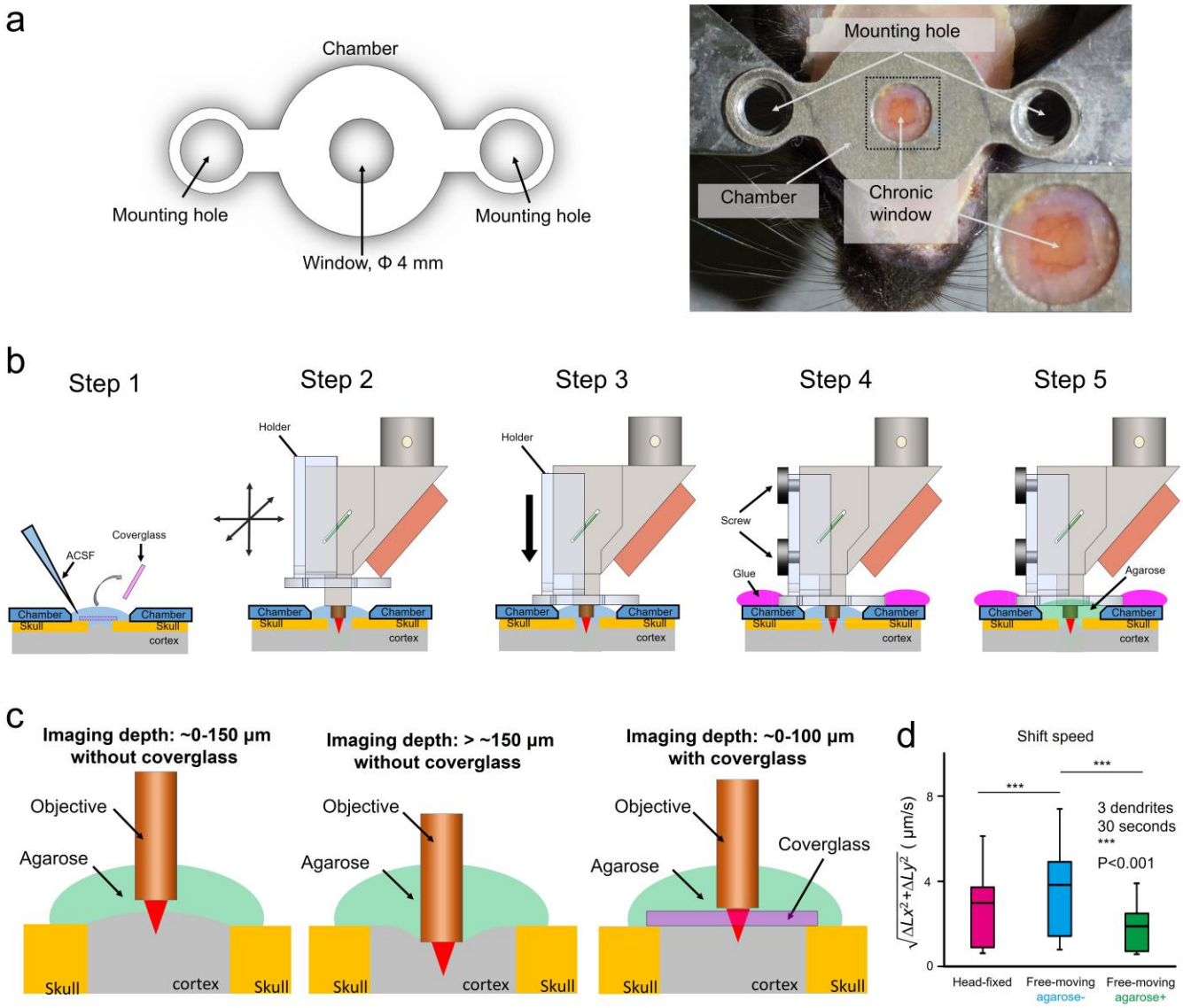
(**a**): 50-s-averaged FHIRM-TPM image of V-1 cortical L2/3 neurons. (**b**): Two-dimensional trajectories of the mouse within the arena (duration: 100 s). Traces were calculated from video records captured by camera-1, and movement speeds are color-coded. (**c**) and (**d**): Time courses (duration: 100 s) of Ca^{2+} transients from three dendrites (circled in **a**) under head-fixed (**c**) and freely-moving conditions (**d**). Time-dependent change in the speed of the mouse is also shown at the bottom. (**e**): Average frequencies of all dendritic Ca^{2+} transients under head-fixed and free-moving conditions. Data are presented as box-and-whisker plot from15 dendrites. Two-tailed paired t-test. *** p <0.001.



Supplementary Figure 7

Preliminary results of the mTPM equipped with another customized miniature optical objective.

Left: Cortical neurons of a mouse were loaded with Cal-520, and imaged with the FHIRM-TPM equipped with another customized miniature objective (NA 0.7, Domilight, China) under head-fixed conditions. A FOV of 170 ×170 mm^2 was achieved. Right: Representative Ca^{2+} transients from selected 4 somata shown in left image. The image speed was 20 Hz and the image depth was 150 mm with a 170-μm-thickness coverslip.



Supplementary Figure 8

How to mount the headpiece onto the head of a mouse.

(**a**) Left, the scheme of the home-made chamber. Right, the photo of a mouse mounted with the chamber and cranial window. (**b**) The step-by-step schematics of how to mount the microscope onto the head of the mouse. Step 1, the cranial window (coverglass) was removed from the skull and a drop of ACSF was placed on the surface of the exposed brain tissue. Step 2, we moved the microscope with a 3-axial motorized stage, and continuously imaged until the interested ROI identified. The holder was pulled up during this stage. Step 3, we moved the holder down to attach to the chamber. Step 4, we injected glue to the edge of the holder to stabilize the holder and the chamber. Next, we fastened two screws to tightly connect the holder to the microscope. Step 5, we put a drop of 1.5% agarose around the end of the objective and the surface of the brain. (**c**) Positions of the objective and the agarose in different conditions. Left: Upon imaging the superficial tissue (0~150 μm) without the coverglass, the gap between the objective and the surface of the brain was filled with the agarose. Middle: Upon imaging the deep tissue (more than 150 μm) without the coverglass, the objective was directly attached the surface of the brain, with the agarose dropped around the objective and the top of the brain. Right: Under the condition of imaging through a coverglass, the agarose was filled in between, and surrounding the objective and the coverglass. Due to the limited working distance of the miniature objective (~200 μm) and the thickness of the coverglass (>100 μm), imaging depth is limited (~0-100 μm for the GRINTECH objective). (**d**) The shift speed of FOV in head-fixed, free-moving conditions with and without the agarose. Data are presented as box-and-whisker plot from 3 dendrites. ***p <0.001.

Supplementary Note 1: Optimizing femtosecond pulse delivery using the HC-920.

During the transmission of ultrafast laser pulses, two factors contribute to the pulse broadening: dispersion by materials, and nonlinear effects at high laser peak powers[1]. In the HC-920 fiber, most of the laser light travels through an air-filled core, thus nonlinear effects are minimal. In the light path of the FHIRM-TPM, light densities at the centers of optical components such as the lenses and mirrors are relative low because of large beam sizes (~2-mm diameter), thus inducing little nonlinearity. Broadening by the miniature components in the front piece of the microscope and the specimen is also negligible due to their relatively short lengths. Therefore, the major broadening effect originates from material dispersion induced by the benchtop optical components and the HC-920 fiber. With simplification, only second-order dispersion is taken into account in our systems[1]. Therefore, the pulse broadening of the whole system B_{system} is:

$$B_{\text{system}} = \left| \sum_n \left\{ L_n \int_{\lambda,\Delta\lambda} D_n(\lambda) \mathrm{d}\lambda \right\} \right| \qquad (1.1)$$

where n represents the number of the optical components in the system; $D_n(\lambda)$ the dispersion parameter induced by the nth component; L_n the optical path length of the nth component; λ and $\Delta\lambda$ are the center wavelength and spectral width of the laser. $|\ |$ is the absolute value of the summation of normal dispersion from lenses (negative $D(\lambda)$) and anomalous dispersion from HC-920 fiber (positive $D(\lambda)$). As $\Delta\lambda << \lambda$ in our system (λ = 920 nm, $\Delta\lambda$ ~15 nm), the equation is simplified into:

$$B_{\text{system}} = \left| \sum_n \left\{ L_n \cdot D_n(\lambda) \cdot \Delta\lambda \right\} \right| \qquad (1.2)$$

And the final pulse width is expressed as:

$$P_{\text{output}} = P_{\text{input}} + \left| B_{\text{system}} \right| \qquad (1.3)$$

where P_{input} and P_{output} denote the input and output pulse widths of the laser. The length of HC-920 used in our system is 1 m ($L_{\text{hc-920}}$); the $D_{\text{hc-920}}$ is measured by the manufacturer and shown in **Fig. 1b**.

Nature Methods: doi:10.1038/nmeth.4305

Therefore, the broadening by HC-920 is:

$$L_{\text{hc-920}} \cdot D_{\text{hc-920}}(920\text{nm}) \cdot \Delta\lambda \sim 1(\text{m}) \cdot 75(\text{ps/nm/km}) \cdot 15(\text{nm}) = 1.125(\text{ps})$$

The lengths of other components such as polarization beam splitters (PBSs), lenses, and AOM (**Supplementary Figs. S5a** and **S5b)** were easily determined; and their $D(\lambda)$ was calculated by the refraction-index information of the glasses they used (such as Scatter N-SF1, N-BAF10, etc. http://www.schott.com/) which is expressed as[1]:

$$GVD = \frac{\lambda^3}{2\pi c^3}\left(\frac{\mathrm{d}^2 n}{\mathrm{d}\lambda^2}\right) \qquad (1.4)$$

and

$$D(\lambda) = -\frac{2\pi c}{\lambda^2} \cdot GVD \qquad (1.5)$$

where GVD is the frequency dependence of the group velocity in a medium, *c* is the speed of light, and *n* is the refractive index of the medium.

Added together, these components induce about -200 fs broadening, so the calculated pulse width output from the HC-920 of an 85-fs transform-limited pulse is about:

(0.085+1.125-0.2) (ps) = 1.01(ps)

which matches the experimentally measured pulse widths (~1 ps) (**Supplementary Fig. S2c)**, and is too broad for efficient two-photon excitation[2]. By inserting a glass tube into the light path with positive dispersion, we pre-compensated the negative dispersion induced by the HC-920. We chose H-ZF62 glass because it has large positive dispersion (~ -458 ps/nm/km in 920 nm, calculated by the refractive-index information supplied by Scott Company and equation (1.4), shown in **Supplementary Fig. S2b**). The length of H-ZF62 needed is calculated as:

(1.125-0.2) (ps)/458(ps/nm/km)/15(nm) = 13.5 (cm);

Considering the small positive dispersion induced by the front piece of the microscope and the

Nature Methods: doi:10.1038/nmeth.4305

specimen, we fabricated a 12-cm-length and 12.7-mm-diameter H-ZF62 tube, smaller than the calculated value. After this "pre-chirping" by H-ZF62 (**Supplementary Fig. S2b** and **Supplementary Fig. S5a)**, the pulse width of the laser coming from the HC-920 fiber was compressed to ~100 fs within the whole range of illumination power tested (**Supplementary Fig. S2d** and **Fig. 1b)**.

Nature Methods: doi:10.1038/nmeth.4305

Supplementary Note 2: Design and theoretical resolution of the FHIRM-TPM.

We followed two guiding rules throughout the optical design of the FHIRM-TPM system, to ensure the imaging quality and to keep the system as compact as possible (**Supplementary Fig. S3**). We chose a high-resolution water-immersion miniature objective (GT-MO-080-018-AC900-450) from GRINTECH GmbH (Jena, Germany); this is finite-corrected with a magnification of 4.76×, an object NA of 0.80, and an image NA of 0.173. It is 6 mm long, 1.4 mm in diameter, and has a working distance of 200 μm. The hybrid lens-assembly consists of two GRIN lenses, one thin Fresnel lens and one plano-convex lens (**Supplementary Fig. S3**). Compared to the objectives used in previous reports[3, 4], the diffraction component (Fresnel lens) inserted into the objective efficiently corrects the large chromatic aberration between excitation and emission wavelengths during two-photon imaging. This results in better beam focusing quality and improved signal collection efficiency[5, 6].

In addition to an achromatic miniature objective, we also used an achromatic collimating lens (#65-286, Edmund Optics Inc., Barrington, NJ, USA; diameter: 2 mm, equivalent focal length: 3 mm, specifically designed for near-infrared light) to collimate the output laser and reduce the chromatic aberration between different frequency components of the femtosecond laser.

A custom-design MEMS scanner (Mirrorcle Technologies Inc., Richmond, CA, USA) was used for fast 2D raster scanning with the first resonant frequency at 6.2 kHz and a scanning angle as large as ±5.0 degrees, which equates to a ±10-degrees optical deflection angle. Compared to other scanning strategies such as fiber scanning with a spiral[7] or Lissajous pattern[8] using piezo actuators, the MEMS scanning is advantageous for its combination of high speed, uniform excitation across the whole FOV, and large scanning angle or FOV. An aspheric lens (#355160B, Lightpath Technologies, Orlando, FL, USA; diameter: 3 mm, equivalent focal length: 2.7 mm) was used as the scan lens to reduce spherical aberration. We custom-made a dichroic mirror (Sunlight Ltd., Fujian, China; size 3 × 3 × 0.2 mm^3, reflection band: 750-1100 nm; transmission band: 400-650 nm) to reflect the laser beam from the scan

Nature Methods: doi:10.1038/nmeth.4305

lens to the objective. All these components were integrated within a ~1 cm^3 aluminum frame (**Fig. 1a**). Finally, we used a custom-made SFB to relay the emission signal from the objective to the remote GaAsP PMT (10770P-40, Hamamatsu, Japan), the total collection efficiency of which is ~80% at 532 nm (**Supplementary Table S2**).

Now we calculate the theoretical resolution of the system. First, the effective excitation NA (NA_{ex}) is determined by the entire light path of excitation. The diameter of the laser beam after the collimating lens D_1 is:

$$D_1 = 2 \cdot NA_{hc\text{-}920} \cdot EFL_c \qquad (2.1)$$

where $NA_{hc\text{-}920}$ is the numerical aperture of the HC-920 (~0.15); and EFL_c the equivalent focal length of the collimating lens (3 mm). Because D_1 (0.9 mm) is larger than the size of the MEMS D_m (0.8 mm for the x-direction and ~0.6 mm for the y-direction, taking account of the obliquity of the MEMS), NA_{ex} is limited by the size of the MEMS and is given by:

$$NA_{ex} = (1/2 \cdot D_m / EFL_s) \cdot NA_o / NA_i \qquad (2.2)$$

where EFL_s is the equivalent focal length of the scan lens (2.7 mm). NA_o, NA_i denote the objective and image NA of the objective, which are 0.8 and 0.173, respectively. Then the maximum excitation NA decided by the x direction of the MEMS is:

$$NA_{ex} = (1/2 \cdot 0.8/2.7) \cdot 0.8/0.173 \sim 0.685$$

Using equations in reference[2] to calculate the diffraction-limited lateral (ω_{xy}) and axial (ω_z) 1/e radii of the point spread function of two-photon excitation (I_{PSF}^2), we have:

$$\omega_{xy} = 0.320 \cdot \lambda / (\sqrt{2} NA_{ex}) \qquad (2.3)$$

$$\omega_z = 0.532 \cdot \lambda / \sqrt{2} \cdot \left[1/(n - \sqrt{n^2 - NA_{ex}^{\ 2}}) \right] \qquad (2.4)$$

Nature Methods: doi:10.1038/nmeth.4305

where n is the refractive index of the immersion medium (~1.34 of normal saline) and λ the wavelength of the excitation laser (~920 nm). For a NA_{ex} of 0.685, $\omega_{xy} = 0.304$ μm and $\omega_z = 1.840$ μm. The resolution is usually defined as the full width at half maximum (FWHM) of the PSF[2], and we have:

$$FWHM_{xy} = 2\sqrt{\ln 2} \cdot \omega_{xy} = 2\sqrt{\ln 2} \cdot 0.304(\mu m) \sim 0.506(\mu m)$$

$$FWHM_z = 2\sqrt{\ln 2} \cdot \omega_z = 2\sqrt{\ln 2} \cdot 1.840(\mu m) \sim 3.063(\mu m)$$

These numbers are close to our measured FWHMs when using 100-nm fluorescent beads ($FWHM_{lateral}$ of 0.66 ± 0.02 μm and $FWHM_{axial}$ of 3.30 ± 0.18 μm, see **Fig. 2**).

Nature Methods: doi:10.1038/nmeth.4305

Supplementary Note 3: A platform to compare benchtop TPM, miniature wide-field fluorescence microscope, and miniature TPM.

We have designed and implemented a platform that is switchable among benchtop TPM, FHIRM-TPM, and miniature wide-field fluorescence imaging configurations while imaging parfocal, concentric FOVs of the same specimen (**Supplementary Fig. S5**. The laser source (Chameleon Vision-S, Coherent, USA; center wavelength: 920 nm) was pre-chirped by a glass tube (**Supplementary Note 1**, followed by an AOM (MT110-B50A1.5-IR-Hk, AA Sa., Orsay Cede, France). The laser was then divided into two beams, one coupled to a 1-m HC-920 used in FHIRM-TPM, and the other coupled to another 1-m HC-920 used in a benchtop TPM (**Supplementary Figs. S5** and **S5**).

In the benchtop mode, the collimated laser from the HC-920 was scanned by a pair of galvo mirrors (6215H, Cambridge Technology, MA, USA), then passing the scan lens (LMS05-BB, Thorlabs Inc, New Jersey, USA), a tube lens (Olympus Japan), and dichroic mirror 1(DM1, DMLP650R, Thorlabs Inc, New Jersey, USA) before finally being delivered to the back focal plane (BFP) of the objective (CFI Apo 40XW NIR, Nikon, Japan; 40× NA 0.8, working distance: 3.5 mm). The fluorescence emission signal from the objective was reflected by DM1, passing a collection lens (LA1213-A, Thorlabs Inc, New Jersey, USA), dichroic mirror 2 (DM2, #87-284, Edmund Optics Inc., Barrington, NJ, USA; reflection band: 375–393, 466–490, and 546–565 nm; transmission band: 420–460, 510–531, 590–624, and 677–725 nm), converged by a condenser (ACL25416U-A, Thorlabs Inc, New Jersey, USA) and finally detected by a GaAsP PMT (7422P-40, Hamamatsu, Japan) (**Supplementary Figs. S5c** and **S5d**). In order to achieve the same excitation NA as that of FHIRM-TPM (0.685), the beam size of the excitation laser at the BFP was set to:

$$2 \cdot 0.685 \cdot 200(\mathrm{mm}) / 40 = 6.85(\mathrm{mm})$$

where 200 mm was the EFL of the matched tube lens for this objective, and 40 the magnification of the objective.

Nature Methods: doi:10.1038/nmeth.4305

In the FHIRM-TPM and miniature wide-field fluorescence microscope configurations (**Supplementary Figs. S5e –S5h**), the miniature objective was attached to a customized tiny three-axial manual micromanipulator (Sigma Koki, Tokyo, JAPAN), which was used to conjugate the image plane of the miniature objective to the focal plane of an air objective (Plan Apo NIR 5X, Mitutoyo, Japan; 5×, NA 0.14, working distance: 37.5 mm) (**Supplementary Fig. S5f** and **Fig. S5g inset**). A 4-port objective lens turret (OT1, Thorlabs Inc, New Jersey, USA) with a high precision of bi-directional repeatability (±2.5 μm) was used to switch between the miniature objective and the 40× Nikon objective. By adjusting the thread of the lens tube before the Nikon objective and the micromanipulator to the miniature objective, we could set the miniature objective and the Nikon objective to have the same focal plane.

In the miniature wide-field fluorescence configuration, a fluorescence illuminator (X-Cite 120Q, Excelitas Technologies, MA, USA) with a blue light filter (MF475-35, Thorlabs Inc, New Jersey, USA; center wavelength: 475 nm, bandwidth: 35 nm) was used as the light source. DM1 in the TPM configuration was changed to dichroic mirror 3 (DM3, #87-284, Edmund Optics Inc., Barrington, NJ, USA; reflection band: 375–393, 466–490, and 546–565 nm; transmission band: 420–460, 510–531, 590–624, and 677–725 nm) to reflect the illumination light to the miniature objective and to transmit the fluorescent emission signal from the miniature objective to a scientific CMOS camera (8050M-GE, Thorlabs Inc, New Jersey, USA) (**Supplementary Fig. S5e** and **Fig. S5f**). Imaging in the benchtop TPM and FHIRM-TPM configurations used similar average powers after the objective, typically 10-25 mW (**Supplementary Table S6**). Owing to the differences in the absorption cross-sections of one-photon and two-photon excitation[9], the average power used in one-photon imaging is usually much lower than that in two-photon imaging[10]. Therefore, we used 100-500 μW illumination (after the miniature objective) in the wide-field fluorescence microscope configuration, similar to the 170-600 μW illumination used in previous reports[11] (**Supplementary Table S6**).

Nature Methods: doi:10.1038/nmeth.4305

Supplementary Table S1. Specifications of HC-920.

HC-920	
MATERIAL	
Fiber material:	Contamination-free silica
Coating material:	Acrylate (single layer)
MANUFACTURER	NKT Photonics A/D
DIMENSIONS	
Core diameter:	8-9 μm
Cladding diameter:	152 μm
Coating diameter:	255 μm
OPTICAL PROPERTIES	
Mode field diameter @ 900 nm:	7 μm
Unpolarized transmission loss @ 920 nm:	100 dB/km
Group velocity dispersion @ 920 nm	75 ps/nm/km
Measured transmission efficiency for 1-m-length @ 920 nm[1]	92 ±2%

[1]:Using Coherent Chameleon Vision-S Ti-sapphire femtosecond laser, center wavelength 920 nm and input power 100 ±1 mW.

Nature Methods: doi:10.1038/nmeth.4305

Supplementary Table S2. Properties of the supple fiber bundle.

Number of fibers	800
Single fiber diameter (µm)	30
Cladding thickness (µm)	~1
Numerical aperture (NA)	0.65
Collection angle (degrees)	80
Transmissivity (%/m)	90
Fill factor (%)	90
Total collection efficiency (%)	80
Manufacturer	Nanjing Chunhui Science and Technology Industrial, China

Nature Methods: doi:10.1038/nmeth.4305

Supplementary Table S3. Comparison of the FHIRM-TPM and the wide-field miniature microscope.

	FHIRM-TPM	**miniature wide field fluorescence microscope[11]**
Resolution	Lateral: 650 nm Axial: 3.3 μm	Lateral: ~2.5 μm (optical 1.5 μm) Axial: None
Maximum field of view	130 × 130 μm^2	700 × 700 μm^2
Maximum speed	40 Hz at 256 × 256 pixels	36 Hz at 640 × 480 pixels
Weight	2.15 g	1.9 g
Penetration depth	~200 μm	Superficial
3D imaging	Capable	Not capable
Multi-color detection	Capable	Not capable
Random access/Line scan	Capable	Not capable
Fine Structure observation	Capable	Not capable
Dense labeling method	Capable	Not capable

Nature Methods: doi:10.1038/nmeth.4305

Supplementary Table S4. Comparison of the FHIRM-TPM and previous mTPMs.

	FHIRM-TPM	**Denk group 2001[8]**	**Helmchen group 2008[7]**	**Kerr group 2009[12]**	**Schnitzer group 2009[13]**
Resolution	**Lateral:** 0.65 μm **Axial:** 3.3 μm	**Lateral:** unknown **Axial:** unknown	**Lateral:** 0.98 μm **Axial:** 7.68 μm	**Lateral:** 0.9 μm **Axial:** unknown	**Lateral:** 1.29 μm **Axial:** 10.3 μm
Maximum field of view	130 × 130 μm^2	65 × 65 μm^2	200 × 200 μm^2	~250 × 250μm^2 *	~100 × 295μm^2
Maximum speed	40 Hz at 256 × 256	2 Hz at 128 × 128	8 Hz at 512 × 512	11 Hz at 64 × 64	4 Hz at 400 × 135
Scanning mechanism	MEMS scan	Fiber scan	Fiber scan	Fiber scan	MEMS scan
Weight	2.15 g	25 g	0.6 g	5.5 g	2.9 g
Excitation wavelength	920 nm	820–850 nm	812 nm	925 nm	790–810 nm
Pulse width after objective	~100 fs	100 fs at 10 mW 1 ps at 180 mW	100 fs	1.8 ps	~110 fs
Laser power after objective	10~25 mW	Not shown	~150-200 mW	100-150 mW	27 mW
Free-moving experiment	Mouse	Rat	Not shown	Rat	Not shown
GCaMP imaging	Capable	Not capable	Not capable	Very low efficiency	Not capable

*: Derived from figures in the paper.

Nature Methods: doi:10.1038/nmeth.4305

Supplementary Table S5. Imaging parameters in text figures.

	Figs. 2a/b	Fig. 3a left	Fig. 3a middle	Fig. 3a right
Mode	FHIRM-TPM	Miniature wide-field	Benchtop TPM	FHIRM-TPM
Sample	100-nm Beads	Neurons (*Thy1*-GFP)	Neurons (*Thy1*-GFP)	Neurons (*Thy1*-GFP)
Imaging volume (xyz) (μm)	50×50×20	250×250×100	80×80×100	80×80×100
Frame size (xyzt)	512×512×100×1 (20 frames/plane)	640×640×100×1 (8 frames/plane)	512×512×100×1 (2 frames/plane)	512×512×100×1 (8 frames/plane)
Frame rate	8 Hz	8 Hz	2 Hz	8 Hz
Laser wavelength	920 nm	470 ±15 nm (lamp)	920 nm	920 nm
Laser power	10 mW	~200 μW (lamp)	10 mW	10 mW
Emission filter	525 ±18 nm	525 ± 18 nm	525 ±18 nm	525 ±18 nm

	Fig. 3b left	Figs. 3b/c middle	Figs. 3b/c right	Fig. 4b
Mode	Miniature wide-field	Benchtop TPM	FHIRM-TPM	FHIRM-TPM
Sample	Neurons (GCaMP-6f)	Neurons (GCaMP-6f)	Neurons (GCaMP-6f)	Neurons (GCaMP-6f)
Imaging FOV (xy) (μm)	250×250	80×80	80×80	130×130
Frame size (xyzt)	640×640×1×4,000 (240 frames averaged)	512×512×1×1,000 (60 frames averaged)	512×512×1×4,000 (240 frames averaged)	256×256×1×4,000 (2000 frames averaged)
Frame rate	8 Hz	2 Hz	8 Hz	40 Hz
Laser wavelength	470 ± 15 nm (lamp)	920 nm	920 nm	920 nm
Laser power after objective	~400 μW (lamp)	15 mW	20 mW	20 mW
Emission filter	525 ±18 nm	525 ± 18 nm	525 ±18 nm	525 ±18 nm

Nature Methods: doi:10.1038/nmeth.4305

	Fig. 4e	**Fig. 4f** up	**Fig. 4f** down
Mode	FHIRM-TPM (line-scan)	FHIRM-TPM	FHIRM-TPM (random access)
Sample	Neurons (GCaMP-6f)	Neurons/astrocytes (Cal-520/SR101)	Neurons/astrocytes (Cal-520/SR101)
Imaging FOV (xy) (µm)	40×1	100×100	20×20
Frame size (xyzt)	256×256×1×200 No average	512×512×1×4,000 (1000 frames averaged)	20×20×4×4,000 (1000 frames averaged)
Frame rate	10 kHz	8 Hz	128 Hz
Laser wavelength	920 nm	920 nm	920 nm
Laser power after objective	15 mW	20 mW	10 mW
Emission filter	52 ±18 nm	525 ± 18 nm 607 ± 18 nm	525 ± 18 nm 607 ± 18 nm

	Fig. S1	**Fig. S6**	**Fig. S7**
Mode	Benchtop TPM (resonant scanner)	FHIRM-TPM	FHIRM-TPM
Sample	Neurons (GCaMP-6f)	Neurons (GCaMP-6f)	Neurons (Cal-520)
Imaging FOV (xy) (µm)	300×300	130×130	170×170
Frame size (xyzt)	512×512×1×4,000 (1800 frames averaged)	256×256×1×4,000 (2000 frames averaged)	256×256×1×4,000 (2000 frames averaged)
Frame rate	30 Hz	40 Hz	20 Hz
Laser wavelength	800/920/1030 nm	920 nm	920 nm
Laser power after objective	15 mW	20 mW	20 mW
Emission filter	525 ± 18 nm	525 ±18 nm	525 ±18 nm

Nature Methods: doi:10.1038/nmeth.4305

Supplementary Table S6. The success rate of free-moving experiments conducted with the microscope.

Paradigms	Total animals	Successes*	mouse ID	Trials	Acquisition time per trial
Freely run	3	3	1	3x	400 s
			2	4x	200 s
			3	3x	200 s
Tail suspension	3	3	2	9x	200 s
			3	5x	200 s
			4	5x	200 s
Step down	2	2	2	3x	400 s
			3	3x	200 s
Social interaction	1	1	2	3x	400s

*: A success refers to an experiment in which we successfully detected neuronal activities from somata, dendrites and/or spines when animals were released and freely behaved. At least three trials at variable intervals and greater than 200s continuous imaging acquisition in each trial were applied to each animal.

Nature Methods: doi:10.1038/nmeth.4305

Supplementary references

1. Agrawal, G.P. in Optics and photonics 1 online resource (xiv, 458 p.) (Academic Press,, San Diego; 2001).
2. Zipfel, W.R., Williams, R.M. & Webb, W.W. Nonlinear magic: multiphoton microscopy in the biosciences. *Nature biotechnology* **21**, 1369-1377 (2003).
3. Rivera, D.R. et al. Compact and flexible raster scanning multiphoton endoscope capable of imaging unstained tissue. *Proceedings of the National Academy of Sciences of the United States of America* **108**, 17598-17603 (2011).
4. Zhang, Y. et al. A compact fiber-optic SHG scanning endomicroscope and its application to visualize cervical remodeling during pregnancy. *Proceedings of the National Academy of Sciences of the United States of America* **109**, 12878-12883 (2012).
5. Matz, G., Messerschmidt, B. & Gross, H. in SPIE Proceedings, Vol. 9304 93041E-93041E-93010 (2015).
6. Matz, G., Messerschmidt, B. & Gross, H. Design and evaluation of new color-corrected rigid endomicroscopic high NA GRIN-objectives with a sub-micron resolution and large field of view. *Optics express* **24**, 10987-11001 (2016).
7. Engelbrecht, C.J., Johnston, R.S., Seibel, E.J. & Helmchen, F. Ultra-compact fiber-optic two-photon microscope for functional fluorescence imaging in vivo. *Optics Express* **16**, 5556 (2008).
8. Helmchen, F., Fee, M.S., Tank, D.W. & Denk, W. A Miniature Head-Mounted Two-Photon Microscope. *Neuron* **31**, 903-912 (2001).
9. Kastrup, L. & Hell, S.W. Absolute optical cross section of individual fluorescent molecules. *Angewandte Chemie* **43**, 6646-6649 (2004).
10. Wang, K. et al. Direct wavefront sensing for high-resolution in vivo imaging in scattering tissue. *Nat Commun* **6**, 7276 (2015).
11. Ghosh, K.K. et al. Miniaturized integration of a fluorescence microscope. *Nature methods* **8**, 871-878 (2011).
12. Sawinski, J. et al. Visually evoked activity in cortical cells imaged in freely moving animals. *Proceedings of the National Academy of Sciences* **106**, 19557-19562 (2009).
13. Piyawattanametha, W. et al. In vivo brain imaging using a portable 2.9 g two-photon microscope based on a microelectromechanical systems scanning mirror. *Optics Letters* **34**, 2309 (2009).

Nature Methods: doi:10.1038/nmeth.4305